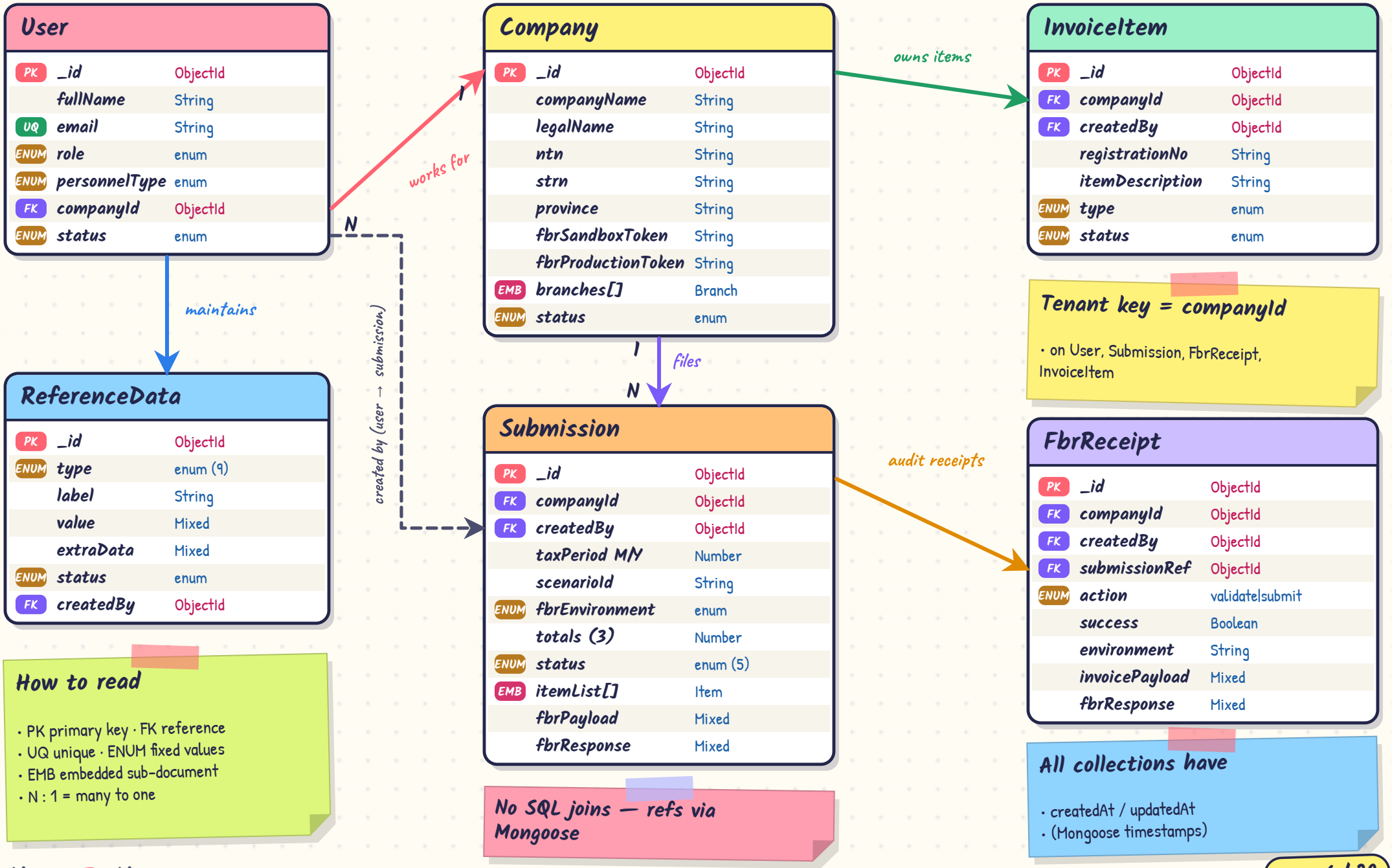
Entity Relationship Map

6 collections · MongoDB



Tenant key = companyId

- on User, Submission, FbrReceipt, InvoiceItem

How to read

- PK primary key · FK reference
- UQ unique · ENUM fixed values
- EMB embedded sub-document
- N : 1 = many to one

No SQL joins — refs via Mongoose

All collections have

- createdAt / updatedAt
- (Mongoose timestamps)

SUBMISSION = ONE INVOICE

header + itemList[]

Submission · header

FK	companyId	ObjectId
FK	createdBy	ObjectId
	submissionType	String
	scenarioId	SN018|SN019
ENUM	fbrEnvironment	sandbox|prod
	taxPeriodMonth	Number
	taxPeriodYear	Number
	invoiceType	String
	invoiceDate	YYYY-MM-DD
	invoiceNumber	String

Submission · totals + state

	totalSaleValue	Number
	totalTaxAmount	Number
	totalBillAmount	Number
ENUM	status	enum (5)
ENUM	validationStatus	enum (3)
	validationErrors	[String]
	fbrPayload	Mixed
	fbrResponse	Mixed
	submittedAt	Date

Status values

- draft → validated
- → ready_for_submission
- → submitted | failed

Submission · parties

	sellerBusinessName	String
	sellerNTN	String
	sellerProvince	String
	sellerAddress	String
	buyerBusinessName	String
	buyerNTN	String
	buyerCNIC	String
	buyerProvince	String
	buyerAddress	String
	buyerType	String

One invoice per submission

- Seller comes from the top level
- Buyer comes from the FIRST item row
- Every row becomes items[]

Calculated on save

- saleValue = qty × unitPrice
- taxAmount = saleValue × taxRate / 100
- totalBill = Σ saleValue + Σ taxAmount

validationStatus

- pending · valid · invalid

itemList[] · price & sale

EMB	itemSNo	Number
	itemDescription	String
	hsCode	String
	uom	String
	quantity	Number
	unitPrice	Number
	saleValue	Number
	taxRate	Number
	taxAmount	Number
	saleType	String
	discount	Number
	totalValueOfSales	Number
	fixedNotifiedValue	Number

Item row ≈ 33 fields

- embedded array — no _id join
- edited as a grid in the UI

Typed carefully

- NTN: 7 alphanumerics
- CNIC: 13 digits
- dates stored YYYY-MM-DD strings
- fbrPayload keeps the exact JSON sent
- fbrResponse keeps bridge envelope

itemList[] · tax, buyer, SRO

	extraTax	Number
	furtherTax	Number
	fedPayable	Number
	stWithheldAtSource	Number
	sroScheduleNo	String
	sroItemSerialNo	String
	buyerNTN / CNIC	String
	buyerBusinessName	String
	buyerType	String
	buyerProvince	String
	sellerProvince	String
	buyerAddress	String
	invoiceType·No·Date	String
	invoiceReferenceNo	String
	reasons · reasonRemarks	String
	petroleumLevyOn	String

Why embed items?

- An invoice is read and written as a whole
- One round-trip loads the form
- Print modals and FBR mapping read one doc
- Totals live beside the rows they sum

Company · User · Branch

tenants and people

Company · FBR registration

registrationNo	String
referenceNo	String
salesTaxRegistered	Boolean
salesTaxRegisteredDate	String
category	String
ppRegIncNo	String
registeredOn	String
taxOffice	String
incomeTaxStatus	String
salesTaxStatus	String

* means required

- Company: name, legalName, ntn, province, address
- User: fullName, email, role

Who is who

- super_admin → no company
- company_user + accountant → creates invoices
- company_user + supervisor → read-only reports

Supervisor sign-in

- Needs a password set by an admin
- ≥ 8 chars, bcrypt hashed
- blank on edit = keep existing

Company · identity & integration

	companyName *	String
	legalName *	String
	ntn *	String
	strn	String
	province *	String
	address *	String
	email · cell	String
	fbrSandboxToken	String
	fbrProductionToken	String
EMB	branches[]	Branch
ENUM	status	active|inactive|suspended
EMB	settings{ }	Object
FK	createdBy	ObjectId

Branch (embedded in Company)

name	String
address	String
principalActivity	String

settings (embedded object)

defaultInvoiceType	String
defaultTaxRate	Number
allowSubmission	Boolean

Company status

- active · inactive · suspended

User

PK	_id	ObjectId
	fullName *	String
UQ	email *	String (lowercase)
	phone	String
	password	String (bcrypt hash)
ENUM	role *	super_admin | company_user
ENUM	personnelType	accountant | supervisor | null
FK	companyId	ObjectId → Company
ENUM	status	active | inactive
	lastLoginAt	Date
FK	createdBy	ObjectId → User

Safety checks at runtime

- inactive user → blocked
- inactive company → blocked
- role + personnel re-read from DB

FBR tokens

- One token per environment
- Production never reuses the sandbox token
- Empty token → bridge default

Personnel sync

- lib/personnel/syncCompanyPersonnel keeps
- company contacts ↔ user accounts aligned

Receipts · Items · Reference Data

audit + lookup lists

FbrReceipt (audit log)

FK	companyId	ObjectId → Company
FK	createdBy	ObjectId → User
FK	submissionRef	ObjectId → Submission
	submissionId	String
ENUM	action	validate | submit
	invoicePayload	Mixed (JSON sent)
	success	Boolean
	mock	Boolean
	environment	sandbox | production
	requestId	String
	httpStatus	Number
	fbrResponse	Mixed
	errorCode	String
	errorMessage	String
	rawEnvelope	Mixed
	receivedAt	Date

rawEnvelope (bridge response)

```
{
"success": true,
"requestId": "...uuid...",
"httpStatus": 200,
"environment": "sandbox",
"mock": false,
"fbrResponse": {
"validationResponse": {...},
"invoiceStatuses": [...] },
"error": { "code", "message" },
"receivedAt": "ISO date"
}
```

InvoiceItem (saved item templates)

FK	companyId	ObjectId → Company
FK	createdBy	ObjectId → User
	registrationNo *	String
	itemDescription *	String
	productDescription	String
	buyerAddress	String
	saleOriginProvinceOfSupplier	String
	destinationOfSupply	String
ENUM	type	buyer type
ENUM	status	active | inactive

index { companyId: 1, status: 1 }

Buyer type (InvoiceItem.type)

- Unregistered · Registered
- Unrecognised · Retail Consumer

Bridge error codes

- FBR_TIMEOUT (35 s default)
- FBR_NETWORK_ERROR
- FBR_UNEXPECTED_RESPONSE
- plus FBR's own status codes

Receipts never lost

- A receipt is written for success and
- failure so there is always an audit trail
- viewable under FBR Invoices

ReferenceData (dropdown values)

ENUM	type	enum (9)
	label *	String
	value *	Mixed
	extraData	Mixed {}
ENUM	status	active | inactive
FK	createdBy	ObjectId → User

ReferenceData.type values

province · invoice_type · buyer_type · tax_rate · hs_code · sale_type · document_type · uom · sro_schedule

Admin-managed lists

- Provinces, HS codes, tax rates, UOM,
- SRO schedules feed the invoice form via
- GET /api/reference-data

getFbrInvoiceStatus()

- Reads status from the envelope:
- status → statusCode → invoiceStatuses[0]
- shown as the badge in FBR Invoices

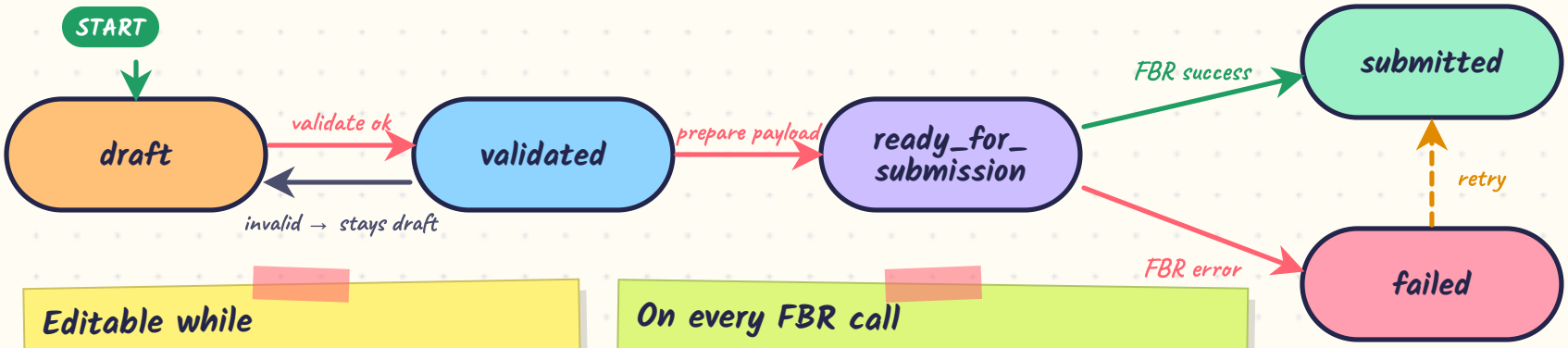
Delete is gated

- Receipt / submission deletes ask for the
- user's password (verify-password)

Status Flow + Data Mapping

life of one submission

Submission.status state machine



Dashboards count

- 5 statuses:
- draft · validated · ready · submitted · failed
- by tax period

Editable while

- status is draft or validated

On every FBR call

- FbrReceipt row + fbrResponse + fbrPayload
- stored; submittedAt only on success

Submission field → FBR invoice payload

sellerNTN	sellerNTNCNIC	item.saleValue	valueSalesExcludingST
buyerNTN | buyerCNIC	buyerNTNCNIC	item.taxAmount	salesTaxApplicable
buyerType	buyerRegistrationType	item.stWithheldAtSource	salesTaxWithheldAtSource
sellerProvince	sellerProvince (FBR name)	item.totalValueOfSales	totalValues
item.itemDescription	productDescription	item.fixedNotifiedValue	fixedNotifiedValueOrRetailPrice
item.uom	uoM	item.invoiceReferenceNo	invoiceRefNo

Type rules in the mapper

- NTN/CNIC are strings
- item.extraTax is a string
- other tax fields are numbers

Display names

- Islamabad → CAPITAL TERRITORY · KPK → KHYBER
- PAKHTUNKHWA · invoice type title-cased

Total bill (modal)

- Σ valueSalesExcludingST + Σ salesTaxApplicable
- (not Σ totalValues — that is PFAD only)